MIL-E-1/200C
29 June 1970
SUPERSEDING
MIL-E-1/200B
31 August 1966

MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, KLYSTRON

## TYPE 6133

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: External cavity, pulsed or CW operation, frequency = 3,525 ±67 MHz

ABSOLUTE RATINGS:

| Parameter: | Ef | Ec1 | Ec2 | Ec3 | Er | Ik | Alt |
|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | Vdc | Vdc | mAdc | ft |
| Maximum: | 6.6 | 325 | 325 | 325 | -300 | 45 | 10,000 |
| Minimum: | 6.0 | --- | --- | --- | -50 | --- | |

PHYSICAL CHARACTERISTICS:

| | |
|---|---|
| Dimensions: | See figure 1 |
| Mounting position: | Any |
| Base: | B4-165 |

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| TEST CONDITIONS: | Ef | Ec1 | Ec2 | Ec3 | Er |
|---|---|---|---|---|---|
| | V | Vdc | Vdc | Vdc | Vdc |
| | 6.3 | 300 | 300 | 300 | -182 to-227 (adjust for max Po) |

GENERAL:

Qualification - Required



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|
| | | | | | MINIMUM | MAXIMUM | |
| | Qualification inspection | | | | | | |
| 1031 | Low-frequency vibration | --- | 10 G; Er = -50 to -300 Vdc | ΔF | --- | 0.1 | MHz |
| 4027 | Temperature coefficient | 4 | t = 20 minutes (max); stabilization time at each temperature extreme | ΔF/° C | --- | 0.15 | MHz |
| | Quality conformance inspection, part 1 | 1 | | | | | |
| 1256 | Electrode current (cathode) | | | Ik | --- | 40 | mAdc |
| 4213 | Reflector voltage | 2 | F = 3,458 to 3,592 MHz | Er | -182 | -227 | Vdc |
| 4229 | Total reflector current | 2 | | Ir | --- | 5.0 | μAdc |
| --- | Fixed frequency | --- | | F | 3,458 | 3,592 | MHz |
| 4250 | Power output | --- | | Po | 95 | --- | mW |
| 4214 | Cathode emission | --- | Ef = 5.7 V; tk = 2 minutes | Ik | --- | 15 | % |
| 4231 | Electronic tuning hysteresis (1) | --- | | --- | --- | 10 | % |
| | Quality conformance inspection, part 2 | | | | | | |
| 1301 | Heater current | --- | Ef = 6.3 V | If | 600 | 720 | mA |
| 1336 | Heater-cathode leakage | --- | | Ihk | 0 | 75 | μAdc |
| 4280 | Electronic tuning range | 5 | Er/50% max Po | ΔF | 20 | --- | MHz |
| --- | Pulsed operation | 3 | | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | |
| --- | Life-test provisions | --- | Group C | t | 500 | --- | hrs |
| --- | Life-test end point | --- | | | | | |
| 4250 | Power output | --- | | Po | 50 | --- | mW |



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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level of II.

2. The tube shall be mounted in a cavity as per Drawing 222-JAN. Er is adjusted for maximum power output into a 50-ohm load with VSWR less than 1.2:1 in the range between -182 and -227 Vdc.

3. Pulsed operation measurement shall be made in the TS-403/U signal generator. The pulse length shall be adjusted to 10 microseconds, and the pulse repetition rate shall be 4, 000 pps. The tube shall pulse smoothly between 1, 760 and 4, 060 MHz with no instability and no shortening of the pulse greater than 2 microseconds. The top of the pulse wave shape viewed on a syncroscope shall be essentially flat, and the pulse amplitude shall not vary by more than 10 dB over the frequency range 1, 760 to 4, 060 MHz, inclusive.

4. Temperature coefficient shall be measured over any 20° C range between 25° C and 80° C with the tube mounted in a cavity as shown on figure 2, or equal. Temperature shall be monitored by means of a thermocouple directly connected to the inner surface of the cavity.

5. The tube shall be mounted in a cavity as per Drawing 222-JAN. Electronic tuning shall be measured in the N = 3-3/4 mode (3-3/4 cycles for complete transit) and the reflector potential at approximately -75 Vdc.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

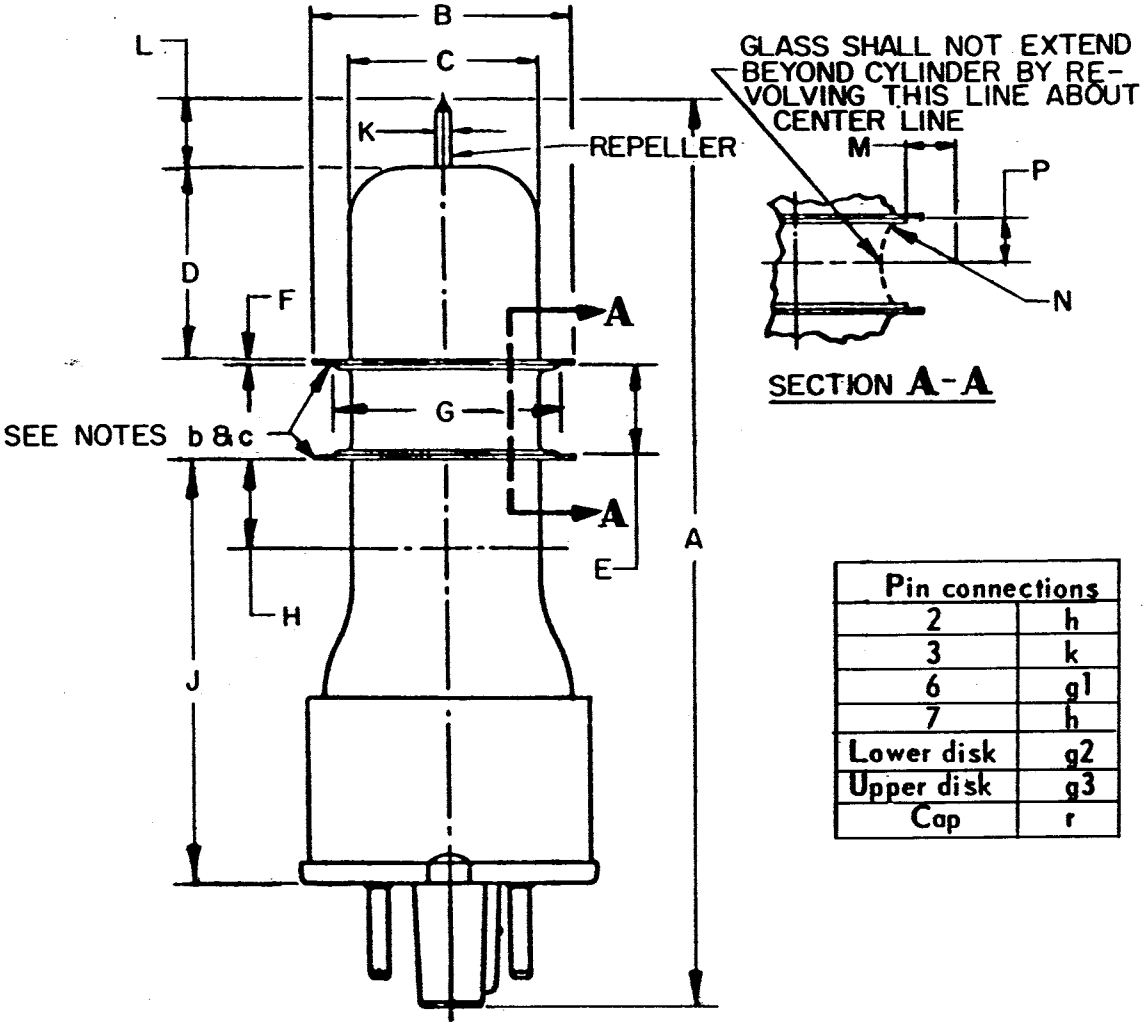
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-1)



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| Pin connections | |
|---|---|
| 2 | h |
| 3 | k |
| 6 | g1 |
| 7 | h |
| Lower disk | g2 |
| Upper disk | g3 |
| Cap | r |

FIGURE 1. Outline drawing of electron tube type 6133.



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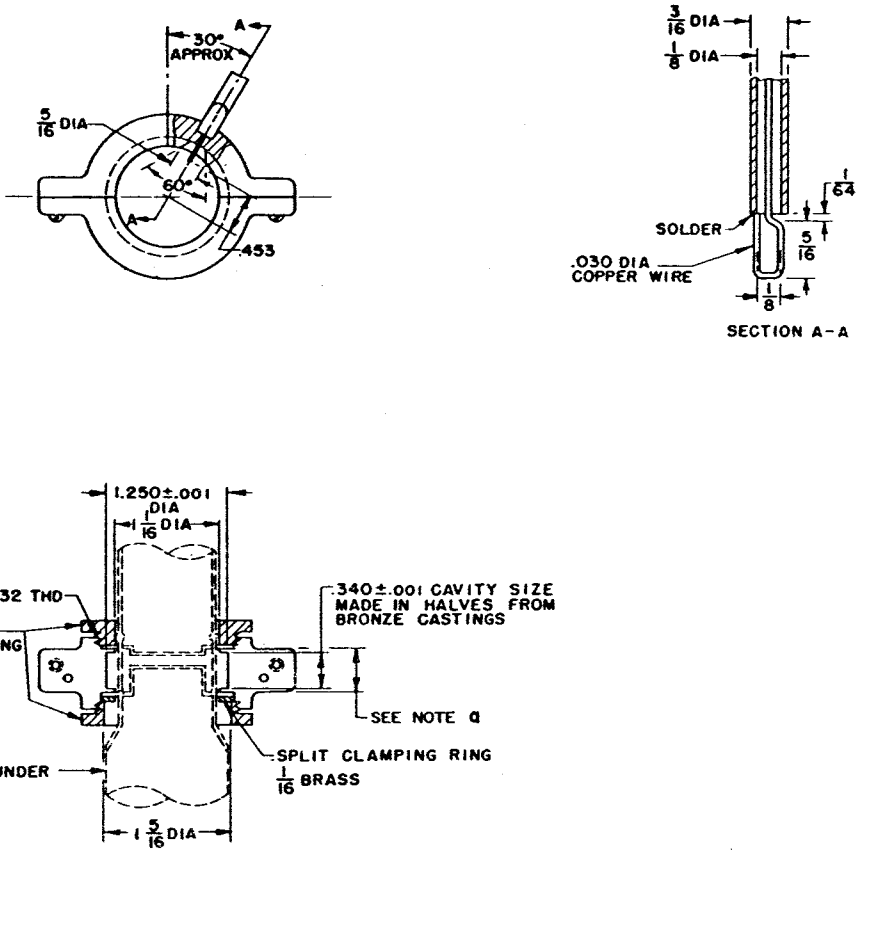
| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | | Notes |
|---|---|---|---|
| | Minimum | Maximum | |
| Quality conformance inspection, part 1 | | | d |
| B | 1.115 (28.32) | 1.135 (28.83) | |
| E | .402 (10.21) | .408 (10.36) | |
| G | | 1.005 (25.53) | a |
| M | | .188 (4.78) | |
| N | .250 (6.35) | | |
| P | .201 (5.11) | .204 (5.18) | |
| Quality conformance inspection, part 2 | | | |
| A | | 4.15 (105.41) | |
| C | .781 (19.84) | .843 (21.41) | |
| D | .797 (20.24) | .891 (22.63) | |
| H | .250 (6.35) | | |
| J | 1.531 (38.89) | 1.781 (45.24) | |
| Quality conformance inspection, part 3 (periodic check) | | | |
| K | .055 (1.40) | .062 (1.57) | |
| L | .313 (7.95) | | |
| F | .010 (.25) Nom | | |

NOTES:
a. Dimension applies to each disk separately.
b. Disks shall be concentric with each other within .015 (.38 mm). The disks shall be concentric within .094 (2.39 mm).
c. Disks shall be smooth, free from tears, and shall be gold plated 20 MSI or 10 MSI nickel plus 10 MSI silver.
d. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level of II.

FIGURE 1. Outline drawing of electron tube type 6133 - Continued.



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NOTES:

a. This dimension, which corresponds to dimension E on figure 1, shall be .401 (10.19 mm) maximum and .396 (10.06 mm) minimum.

b. Cavity and loop to be silver plated 50 MSI per QQ-S-365, other parts NICI chrome flashed.

FIGURE 2. Cavity.



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